

The Mathematics of MOS / TATIANA

A Cellular–Sheaf Cognitive Architecture over a
Stratified Simplicial Complex

An implementation-free mathematical description

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July 24, 2026

Abstract

We give a complete, implementation-free mathematical description of the cognitive architecture called TATIANA (its state object is the *Modular Operating System*, MOS). The organising object is a *cognitive state space* $\mathcal{M} = (C, F, s)$: a two-level (stratified) simplicial complex C , a cellular sheaf F assigning geometric data to each piece of C , and a global section s representing the system’s current coherent state. The architecture wraps a fixed large language model (LLM) used as a black-box reasoning oracle; its mathematical content is not the reasoning itself but the *organisation, measurement, and evolution* of whole reasoning acts. We define every object from first principles, derive the two quantities the system actually computes at every step—the sheaf Laplacian discord ω and its normalised, provably bounded form $\rho \in [0, 1]$ —and record every “minute computation” the engine performs: the precision-weighted Gaussian coarse-graining π_v , the Bures–Wasserstein concept distance, the Woodbury concept-fusion, the Anderson–Morley normaliser, the cohomological diagnostics H^0, H^1, b_0 , and the growth operators {bind, collapse, split, merge} together with the operator-algebra growth axis $\mathfrak{D}(t) \rightarrow \mathfrak{D}(t+1)$. Throughout, boxed *claims-and-non-claims* remarks state precisely what each construction does and does not establish; self-honesty is the single invariant the whole design exists to protect.

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1 Introduction: the object and the thesis

1.1 What is being described

Intuition. Picture a research assistant that is not one mind but a small society of specialist organs—one that remembers, one that searches, one that checks, one that reasons. MOS is the mathematics of *that society*: who is bound to whom, whether their beliefs agree, by how much they disagree and exactly where, and how the society rewires itself as it works. The individual reasoning is done by an external LLM; the mathematics here governs the *society*, not the neurons.

Definition 1.1 (Cognitive state space). A *cognitive state space* is a triple $\mathcal{M} = (C, F, s)$ where

- (i) C is a stratified (two-level) finite simplicial complex (Section 2);
- (ii) F is a cellular sheaf of finite-dimensional real inner-product spaces on C (Section 3);
- (iii) s is a distinguished 0-cochain of F , the *current state*, whose coherence is measured by the sheaf Laplacian (Section 4).

The space evolves in discrete time through structural operators drawn from an operator algebra $\mathfrak{D}$ that is *itself* allowed to grow (Section 9).

1.2 The thesis, stated plainly

The architecture makes one honest bet and refuses one seductive overclaim.

Claims and non-claims. The bet (defensible): a persistent, self-checking, self-restructuring *organiser* wrapped around a fixed LLM can, over time, produce better-grounded and internally-consistent research artifacts than the raw LLM, through *measured cross-organ error-correction* and *accumulation*. The advantage is *architectural*, expressed as operations on the single object $\mathcal{M}$. **The refusal:** no simplex proves a theorem and no Laplacian integrates a differential form. The per-inference reasoning quality is bounded by the LLM. The mathematics here does not raise that ceiling; it organises, measures, and error-corrects *around* it. Section 14 makes this boundary precise and states where it can, and cannot, be pushed.

1.3 How to read this document

Each central object is presented in three layers, so the same text serves a high-school visitor and a topologist: (a) a boxed **Intuition** in plain language; (b) a formal **Definition/Construction**; (c) the exact **computation** the engine performs, with derivations. Sections 2–4 are the conceptual spine. Sections 5–6 and 9–12 record the minute computations. Section 14 is the honest boundary. Appendix A is the driving application (a thermally-weighted Hodge–de Rham spectrum on $\mathbb{CP}^3$).

2 The base complex C

2.1 Simplicial complexes and n -ary binding

Intuition. A graph records *pairwise* facts (“ A relates to B ”). But a theorem that depends on three hypotheses is *one* three-way fact, not three separate pairs. A simplicial complex is the smallest honest language for facts of any arity: a filled triangle is a genuine three-way binding, not three edges.

Definition 2.1 (Abstract simplicial complex). Let V be a finite vertex set. An *abstract simplicial complex* C on V is a family of nonempty subsets of V (the *simplices*) that is *downward closed*: if $\sigma \in C$ and $\emptyset \neq \tau \subseteq \sigma$ then $\tau \in C$. A simplex of cardinality $k + 1$ is a k -*simplex*; its $(k - 1)$ -subsets are its *faces*. Thus 0-simplices are vertices, 1-simplices edges, 2-simplices filled triangles, and so on.

Remark 2.2 (Simplices are n -ary relations). A k -simplex $\{v_0, \dots, v_k\}$ models a single $(k+1)$ -ary binding among its vertices. In the engine this is exactly the stored object `insert_n_ary_relation`: a fact with $k+1$ constituents is one simplex, and downward closure is the requirement that every sub-collection of a jointly-bound fact is itself recorded. This earns the later use of b_1 (Section 8): a 1-cycle is a real structural hole only because a filled 2-simplex genuinely means “these three are jointly bound,” not merely pairwise.

2.2 The two-level stratification

Intuition. The society has two scales. Zoom out and the vertices are whole *organs* (Search, Memory, Reasoning, Verify, ...). Zoom into one organ and it is itself a little complex whose vertices are *concepts*. The same geometry runs at both scales; the two are glued by a coarse-

graining map (Section 5).

Construction 2.3 (The stratified complex). C carries two strata.

- The *coarse complex* K has as vertices the cognitive organs

$$\mathcal{V}(K) = \{\text{PLANNER, SEARCH, CANONICALIZER, CURATOR, WORKINGMEMORY, REASONING, VERIFY}\}.$$

A simplex $\sigma \in K$ is a *bound coalition* of organs acting as one unit; it carries a time-dependent *coupling weight* $w(\sigma, t) \in [0, 1]$.

- For each organ v , a *fine complex* C_v has concepts as vertices and n -ary conceptual bindings as simplices; this is the organ’s internal knowledge.

The stratification resolves an early ambiguity (“are the vertices concepts or subsystems?”) by answering *both*, at two levels.

Remark 2.4 (Two structures on the same vertices). A logical prerequisite chain is *directed* (A is needed for B); a simplicial complex is undirected. The engine therefore keeps two relations on the same vertices: a directed poset for logical dependency and the (undirected) simplices for n -ary co-binding. They are never conflated.

3 The sheaf F

Intuition. On top of the shape we lay *data*. Each organ holds a vector describing what it currently believes; each binding holds a space in which two organs’ beliefs are meant to be compared. A sheaf is exactly this: local data on every piece, plus rules for comparing across pieces. A “coherent thought” is a choice of data that agrees everywhere it is compared.

Definition 3.1 (Cellular sheaf, after Curry and Hansen–Ghrist). A *cellular sheaf* F of real inner-product spaces on C assigns

- to each cell σ a finite-dimensional inner-product space $F(\sigma)$, the *stalk*;
- to each incident face pair $\sigma \trianglelefteq \tau$ (i.e. σ a face of τ) a linear *restriction map* $F_{\sigma \trianglelefteq \tau} : F(\sigma) \rightarrow F(\tau)$,

compatibly on chains of incidences. In this project all vertex and edge stalks are $\mathbb{R}^d$ with $d = 384$ (the embedding dimension, Section 6); the restriction maps of the coarse sheaf are indexed $F_{v \trianglelefteq e}$ for a vertex v on edge e .

Definition 3.2 (Cochains and sections). A *0-cochain* is a choice $x = (x_v)_v$ with $x_v \in F(v)$; write $C^0(C; F) = \bigoplus_v F(v)$. Likewise $C^1(C; F) = \bigoplus_e F(e)$. A *global section* is a 0-cochain whose organ beliefs agree under every restriction (Definition 4.1). The current state s of Definition 1.1 is a 0-cochain; it is a genuine “thought” precisely when it is a global section.

Remark 3.3 (The stalk data is a real embedding, never a placeholder). Stalk vectors are sentence-embeddings from a fixed local encoder (dimension $d = 384$). Two design invariants are enforced as *type-level* facts, because their violation silently corrupts every downstream distance: (i) dimensions never truncate or pad—a mismatch raises; (ii) an idle organ has *no* stalk value (a formal $\perp$), never the zero vector, because $0 \in \mathbb{R}^d$ is a real point that would drag every distance toward the origin and fabricate a measurement. These are the sheaf-theoretic content of “do not fake data.”

4 Coherence: the sheaf Laplacian and the score ρ

This section derives the two numbers the engine computes on *every* tick.

4.1 Coboundary and Laplacian

Definition 4.1 (Coboundary). Fix an arbitrary orientation of each edge $e = \{u, v\}$. The *coboundary* $\delta : C^0(C; F) \rightarrow C^1(C; F)$ acts edgewise by

$$(\delta x)_e = F_{u \leq e}(x_u) - F_{v \leq e}(x_v) \in F(e). \quad (1)$$

A 0-cochain x is a *global section* iff $\delta x = 0$, i.e. every bound pair agrees after mapping into the shared comparison space $F(e)$.

Definition 4.2 (Sheaf Laplacian and discord). The (degree-0) *sheaf Laplacian* is

$$L = \delta^\top \delta : C^0 \rightarrow C^0, \quad (2)$$

a symmetric positive-semidefinite operator. The *discord* of a state x is the Dirichlet energy

$$\omega(x) = x^\top Lx = \|\delta x\|^2 = \sum_{e=\{u,v\}} \|F_{u \leq e}(x_u) - F_{v \leq e}(x_v)\|^2 \geq 0. \quad (3)$$

Proposition 4.3 (A coherent thought is a kernel element). $\omega(x) = 0 \iff \delta x = 0 \iff x \in \ker L$. Moreover the space of global sections is exactly the zeroth sheaf cohomology,

$$H^0(C; F) = \ker \delta = \ker L. \quad (4)$$

Proof. $\omega(x) = \|\delta x\|^2 = 0 \iff \delta x = 0$ since $\|\cdot\|$ is a norm. As $L = \delta^\top \delta$, $Lx = 0 \Rightarrow x^\top Lx = \|\delta x\|^2 = 0 \Rightarrow \delta x = 0$; conversely $\delta x = 0 \Rightarrow Lx = 0$. Finally $H^0 = \ker \delta^0$ because there are no (-1) -cochains to quotient by. $\square$

Intuition. ω is a single number: the total squared disagreement across all bindings. It is zero exactly when the society holds one coherent thought. And “a coherent thought” is now a precise mathematical object—a vector in the kernel of a matrix—not a mood.

4.2 The identity-restriction baseline

Proposition 4.4 (Graph-Laplacian form). If every $F_{v \leq e} = \text{Id}_{\mathbb{R}^d}$, then $\omega(x) = \sum_{e=\{u,v\}} \|x_u - x_v\|^2$ and $L = L_K \otimes I_d$, where L_K is the ordinary combinatorial graph Laplacian of the coarse 1-skeleton. Consequently the spectrum of L is that of L_K with each eigenvalue repeated d times.

Claims and non-claims. The current engine uses identity restriction maps (“do bound organs agree in embedding space?”). This is honest and cheap but it is the *crudest* sheaf: with a connected 1-skeleton, $\ker L$ is the diagonal $\{x : x_u = x_v \ \forall u, v\}$, so the only coherent states are *consensus*, and reconciliation is plain averaging. The “quality of a connection” lives in *non-trivial* restriction maps; promoting them (learned $F_{v \leq e}$, e.g. by closed-form Procrustes/CCA alignment) is the identified route from a diagnostic sheaf to a computational one. This is stated as a direction, not as an implemented result (Section 14).

4.3 The normalised score $\rho \in [0, 1]$ and its bound

Intuition. ω has no natural scale—bigger complexes have bigger ω . We want a dimensionless thermometer in $[0, 1]$: 0 = perfect agreement, 1 = maximal possible disagreement. Crucially we get it *without* an eigendecomposition, using a classical bound on the largest Laplacian eigenvalue.

Definition 4.5 (Coherence score). Let $X \in \mathbb{R}^{|V| \times d}$ stack the active stalks as rows, with Frobenius norm $\|X\|_F$. With the *Anderson–Morley normaliser*

$$B = \max_{e=\{u,v\} \in E} (\deg u + \deg v), \quad (5)$$

the *coherence score* (in the identity-restriction regime) is

$$\rho = \frac{\omega}{B \|X\|_F^2} \in [0, 1]. \quad (6)$$

When $E = \emptyset$ or $\|X\|_F = 0$, ρ is *undefined* ($\perp$, “unknown”), never 0 and never 1.

Proposition 4.6 ($\rho \in [0, 1]$). *With identity restrictions, $0 \leq \rho \leq 1$.*

Proof. Lower bound: $\omega \geq 0$ and $B \|X\|_F^2 > 0$. Upper bound: by the Anderson–Morley inequality [4], the largest eigenvalue of L_K satisfies $\lambda_{\max}(L_K) \leq \max_{\{u,v\} \in E} (\deg u + \deg v) = B$. By Proposition 4.4, $\lambda_{\max}(L) = \lambda_{\max}(L_K) \leq B$, hence $\omega = x^\top Lx \leq \lambda_{\max}(L) \|x\|^2 \leq B \|X\|_F^2$, where $\|x\| = \|X\|_F$ under the row-stacking identification. Dividing gives $\rho \leq 1$. $\square$

Remark 4.7 (Per-edge diagnostics and fragmentation). Because (3) is a *sum over edges*, ω decomposes into per-edge terms $\omega_e = \|(\delta x)_e\|^2$. The *worst edge* $\arg \max_e \omega_e$ names the guilty coalition, so remediation can be *aimed* rather than global—provided $\omega > \text{tol}$ (below tolerance the $\arg \max$ is meaningless and is returned as $\perp$). Separately, the number of connected components b_0 of the active 1-skeleton is computed by union-find; $b_0 > 1$ flags that any coherence claim is only *local* to a component. Both are $O(|E|)$.

Remark 4.8 (Spectral decomposition, when wanted). Because L is genuinely symmetric PSD, the spectral theorem applies exactly (not by analogy): eigenpairs (λ_i, ϕ_i) of L are honest *modes of disagreement*—small λ are droppable noise, large λ structural conflict. This is the rigorous replacement for the earlier, unjustified “cognitive FFT.” It is used only for diagnostics; the fast path (6) needs no eigensolver.

5 The coarse-graining π_v : precision-weighted fusion

Intuition. An organ’s single coarse belief-vector must summarise the many concepts living inside it. Not all concepts are equally certain, so we do not average them equally: a near-certain concept should dominate. The right notion of “weight” is *precision* (inverse variance), and the fusion is exactly how independent noisy measurements of one quantity are optimally combined.

Definition 5.1 (Concept as a Gaussian). A concept is modelled as a Gaussian $\mathcal{N}(\mu_i, \Sigma_i)$ on $\mathbb{R}^d$ with a low-rank-plus-isotropic covariance

$$\Sigma_i = U_i U_i^\top + D_i I_d, \quad U_i \in \mathbb{R}^{d \times k_i}, \quad D_i > 0. \quad (7)$$

μ_i is the embedding; $U_i U_i^\top$ is anisotropic “learned spread”; D_i is an isotropic noise floor (the epistemic humility of the concept).

Construction 5.2 (The coarse-graining map π_v). For organ v with active concepts $\{\mathcal{N}(\mu_i, \Sigma_i)\}_{i=1}^n$, the coarse stalk $\pi_v \in F(v) = \mathbb{R}^d$ is the maximum-likelihood fusion of the concepts regarded as independent estimates of one latent “what this organ believes”:

$$\pi_v = \left(\sum_{i=1}^n \Sigma_i^{-1} \right)^{-1} \left(\sum_{i=1}^n \Sigma_i^{-1} \mu_i \right). \quad (8)$$

Each concept pulls π_v toward itself in proportion to its precision Σ_i^{-1} . This is the n -way, order-independent, non-destructive generalisation of the pairwise fusion of Section 6. If v has no active concept, $\pi_v = \perp$ (idle organs are excluded from ρ , Remark 3.3).

Proposition 5.3 (Why (8) is the fusion). *If θ is a latent vector and each concept is an independent observation $\mu_i = \theta + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0, \Sigma_i)$, then the log-likelihood is $-\frac{1}{2} \sum_i (\mu_i - \theta)^\top \Sigma_i^{-1} (\mu_i - \theta) + \text{const}$; setting its θ -gradient to zero gives $(\sum_i \Sigma_i^{-1})\theta = \sum_i \Sigma_i^{-1} \mu_i$, i.e. (8). Order-independence is manifest (both sums are commutative). Non-destructiveness means no concept is discarded: all contribute to both the precision sum and the mean.*

The two computational paths (both exact).

- **Isotropic fast path.** When all U_i are empty (the present reality of a freshly grown concept), $\Sigma_i^{-1} = D_i^{-1}I$ and (8) collapses to the scalar precision-weighted average

$$\pi_v = \frac{\sum_i D_i^{-1} \mu_i}{\sum_i D_i^{-1}}, \quad (9)$$

in $O(nd)$ with no matrix inversion.

- **General path.** With $U_i \neq \emptyset$, invert each Σ_i by the Sherman–Morrison–Woodbury identity

$$\Sigma_i^{-1} = D_i^{-1}I - D_i^{-1}U_i(I_{k_i} + D_i^{-1}U_i^\top U_i)^{-1}U_i^\top D_i^{-1}, \quad (10)$$

accumulate into a dense information matrix, and solve once: $O(ndk + d^3)$. A floor $D_i \geq 10^{-6}$ prevents a near-certain axiom from producing $1/D_i \approx 10^{16}$ and a numerical blow-up; it pins π_v strongly but finitely.

Remark 5.4 (Verified exactness). The general path was checked numerically against a dense $(\sum_i \Sigma_i^{-1})^{-1}$ solve and agreed to 2.2×10^{-16} (machine precision); equal-precision inputs reduce to the plain mean and a 9:1 precision ratio pulls π_v to the certain concept, as required. This is the two-level glue: after execution, each organ’s coarse stalk is *overwritten* by its own π_v , so coarse ρ and fine geometry become one measurement at two resolutions.

6 Semantic geometry: distance, fusion, entropy

The concepts of Definition 5.1 form a geometry of Gaussians. Three computations act on it.

6.1 Bures–Wasserstein distance between concepts

Intuition. How far apart are two *uncertain* concepts? Not just their centres—their spreads matter too. The 2-Wasserstein (“optimal transport”) distance between Gaussians answers this and is a true metric.

Definition 6.1 (2-Wasserstein between Gaussians).

$$W_2^2(\mathcal{N}(\mu_1, \Sigma_1), \mathcal{N}(\mu_2, \Sigma_2)) = \|\mu_1 - \mu_2\|^2 + \text{Tr } \Sigma_1 + \text{Tr } \Sigma_2 - 2 \text{Tr} \left[(\Sigma_1^{1/2} \Sigma_2 \Sigma_1^{1/2})^{1/2} \right]. \quad (11)$$

Proposition 6.2 (Isotropic reduction to $O(k^3)$). *If $\Sigma_1 = D_1 I$ is isotropic and $\Sigma_2 = U_2 U_2^\top + D_2 I$ with $U_2 \in \mathbb{R}^{d \times k}$, then, writing $\{\sigma_j^2\}_{j=1}^k$ for the eigenvalues of the $k \times k$ matrix $U_2^\top U_2$,*

$$W_2^2 = \|\mu_1 - \mu_2\|^2 + (\text{Tr } \Sigma_1 + \text{Tr } \Sigma_2) - 2\sqrt{D_1} \left[\sum_{j=1}^k \sqrt{\sigma_j^2 + D_2} + (d - k)\sqrt{D_2} \right]. \quad (12)$$

Proof. $\Sigma_1^{1/2} = \sqrt{D_1} I$, so $\Sigma_1^{1/2} \Sigma_2 \Sigma_1^{1/2} = D_1 \Sigma_2$ and $(D_1 \Sigma_2)^{1/2} = \sqrt{D_1} \Sigma_2^{1/2}$, giving the cross term $2\sqrt{D_1} \text{Tr } \Sigma_2^{1/2}$. The eigenvalues of $\Sigma_2 = U_2 U_2^\top + D_2 I$ are $\sigma_j^2 + D_2$ (on $\text{im } U_2$, $j = 1, \dots, k$) and D_2 (with multiplicity $d - k$ on $\ker U_2^\top$); the nonzero eigenvalues of $U_2 U_2^\top$ equal those of $U_2^\top U_2$. Summing $\sqrt{\lambda}$ gives (12). Only a $k \times k$ eigendecomposition is needed. $\square$

Remark 6.3. This is exact for isotropic Σ_1 , not a diagonal approximation. It is the distance the engine uses to decide whether two stored concepts are near enough to merge, and it feeds the acquisition/consolidation decisions of Section 9.

6.2 Pairwise fusion and its entropy

Definition 6.4 (Precision-weighted fusion). The fusion of two concepts is the information-form product

$$\Lambda_i = \Sigma_i^{-1}, \quad \Sigma_\star = (\Lambda_1 + \Lambda_2)^{-1}, \quad \mu_\star = \Sigma_\star(\Lambda_1 \mu_1 + \Lambda_2 \mu_2), \quad (13)$$

i.e. (8) for $n = 2$. Full covariances are fused via Woodbury (10), not element-wise.

Proposition 6.5 (Fused entropy by the matrix determinant lemma). *The differential entropy $h = \frac{1}{2} \ln((2\pi e)^d \det \Sigma_\star)$ is computed from a $k \times k$ determinant: for $\Sigma = U U^\top + D I$,*

$$\det \Sigma = \det(DI + U U^\top) = D^d \det(I_k + D^{-1} U^\top U). \quad (14)$$

Proof. Matrix determinant lemma: $\det(A + U V^\top) = \det A \det(I + V^\top A^{-1} U)$ with $A = D I$, $V = U$. Then $\det(DI) = D^d$ and $V^\top A^{-1} U = D^{-1} U^\top U$. $\square$

Remark 6.6 (Uncalibrated confidence). The noise floor D of a *grown* concept is set from the generating LLM’s self-reported confidence c via $D = -\ln c$ (so $c \rightarrow 1 \Rightarrow D \rightarrow 0$). Against the deployed model, token log-probabilities are unavailable (the provider returns a hard error when they are requested), so c is genuinely unknown; the engine then uses a named, warned prior $D = \text{UNCALIBRATED_VARIANCE_PRIOR} = 1.0$ rather than fabricating a value. This is why the coherence score ρ —not self-reported confidence—is the primary signal.

7 The global section and the fine dynamics

Intuition. Inside an organ, reasoning is a *flow*: new material perturbs the internal state, and a gradient descent pulls it back toward internal coherence, subject to hard constraints (axioms that must hold) and soft constraints (inferred, bendable).

Construction 7.1 (Fine state and deductive flow). The fine state of an organ is a configuration $x \in (\mathbb{R}^d)^m$ over its m active concepts, with a scalar *conflict functional* $\Phi(x) \geq 0$ measuring internal inconsistency (the fine-level analogue of ω). Reasoning integrates the gradient flow

$$\dot{x} = -\nabla \Phi(x) \quad (15)$$

by a fixed-step fourth-order Runge–Kutta (RK4) scheme. The reported δ -strain of a step is the achieved change $\Delta\Phi$. Two constraint types shape Φ : *rigid* constraints (user-stated axioms) act as hard orthogonal projections onto an affine subspace via a projector P ; *fluid* constraints (inferred facts) act as soft quadratic penalties.

Remark 7.2 (Observed sign behaviour). On a real run, injecting a new concept raised Φ by a structural jump (δ -strain = +0.202) and the subsequent RK4 flow pulled it back down (δ -strain = −0.081): a structural expansion followed by a deductive relaxation, exactly the intended dynamics.

Claims and non-claims. The precise closed form of Φ is an implementation-level choice (a penalised quadratic on the fine complex) and is deliberately not over-specified here; what is mathematically committed is (15)—a gradient flow on a nonnegative conflict functional—and the rigid/fluid split as hard-projection vs soft-penalty. Earlier drafts posited an infinite-dimensional Banach state and a Fréchet derivative of the whole pipeline; both were *removed* as unjustified—the pipeline runs through discrete token sampling and database lookups and is nowhere differentiable (see Section 8.2).

8 Cohomology and the diagnostic hierarchy

8.1 H^0 , H^1 and the b_1 distinction

Definition 8.1 (Sheaf cohomology, low degrees). With coboundary $\delta : C^0 \rightarrow C^1$ (and higher coboundaries on the full complex),

$$H^0(C; F) = \ker \delta^0 \quad (\text{global sections}), \quad H^1(C; F) = \ker \delta^1 / \text{im } \delta^0 \quad (\text{gluing failure}).$$

Intuition. Two very different pathologies must never be confused. A *structural hole* ($b_1 \neq 0$, a Betti number of the bare shape) means a *binding is missing*—no theorem yet ties some loop together. A *gluing failure* ($H^1 \neq 0$ of the sheaf) means the data are *contradictory*—the organs cannot be reconciled. The first calls for *growth*; the second calls for *repair*. Treating a contradiction as a hole would make you add structure when you should resolve a conflict.

Remark 8.2 (b_1 is earned here). b_1 is a defensible diagnostic *only* because simplices were defined to mean genuine n -ary binding (Remark 2.2); a filled 2-simplex really asserts a joint relation, so an unfilled 1-cycle really is a missing joint relation. This is what most “topological data analysis on knowledge graphs” fails to justify.

8.2 The uncertainty stack

Intuition. Coherence is not correctness. Four instruments, ordered by cost, answer four different questions; they are not interchangeable.

Instrument	Question	Cost	When
ρ (coherence)	Are my parts agreeing?	$\approx$ free	every tick
$\hat{\rho}$ vs ρ	Did I <i>know</i> whether I’d agree?	free (k -NN)	once history exists
b_0, b_1	What is structurally missing/broken?	medium	on conflict, active subcomplex
perturbation J	Am I stable to paraphrase?	$k \times$ LLM calls	high-stakes only
VERIFYOP	Am I actually <i>right</i> ?	medium	promotion gate

Remark 8.3 (Perturbation is variance, not a derivative). Stability is measured as *paraphrase-perturbation variance*: run k semantically equivalent paraphrases and measure the spread of outcomes. This replaces an earlier “Fréchet derivative $\|\partial F/\partial x\|$ ” that does not exist, because F factors through discrete LLM sampling, a DAG generator, and database lookups—nothing differentiable. The cost is $k \times$ API calls and must be rationed.

Remark 8.4 (Expected precision, and the feedback edge). $\hat{\rho}$ is a similarity-weighted k -NN prediction of ρ from a log of past (query embedding, ρ) pairs. The signed gap $\hat{\rho} - \rho$ separates *competence*, *known ignorance* (“confidently I do not know this”), and *hallucination risk* (predicted agreement, observed disagreement). A VERIFYOP failure updates this store, so self-knowledge improves with use.

9 Dynamics: growth operators and the algebra $\mathfrak{D}$

Intuition. Learning *is* change of shape. The complex binds organs that work together, prunes idle ones, and—when a procedure repeatedly succeeds—promotes it to a new first-class operator. Four axes of growth, one of which grows the very set of moves the system can make.

Definition 9.1 (Structural operators). The base moves on C are the typed operators {bind, collapse, split, merge} each logged to an immutable mutation history. (The earlier terminology “Pachner move” was discarded: a knowledge complex is not a manifold, so the Pachner invariance guarantees do not transfer.)

Construction 9.2 (The four growth axes). Evolution proceeds along

$$\underbrace{\delta C_{\text{fine}}}_{\text{concepts (semantic memory)}}, \quad \underbrace{\delta C_{\text{coarse}}}_{\text{coalitions (functional connectivity)}}, \quad \underbrace{\delta F}_{\text{stalk/restriction data}}, \quad \underbrace{\delta \mathfrak{D}}_{\text{new operators (procedural skill)}}.$$

The fourth axis is the essential one: the operator algebra is time-dependent, $\mathfrak{D}(t) \rightarrow \mathfrak{D}(t+1)$.

Definition 9.3 (Hebbian coupling and decay). For a coalition σ with weight $w(\sigma, t) \in [0, 1]$: co-activation increments w (“fire together $\Rightarrow$ wire together”); idle steps apply $w(\sigma, t+1) = (1 - \gamma)w(\sigma, t)$ for a decay rate γ , and a coalition falling below a bind threshold *collapses* (garbage collection). A pair past the bind threshold is a 1-simplex (edge) of K .

Construction 9.4 (Consolidation via persistence, and microneurons). Over the weight filtration, *persistent homology* reads which coalitions and concept-cycles *survive*; survivors are promoted into permanent structure. In particular a *microneuron* is a promoted operadic sub-tree: when a composite DAG repeatedly resolves conflict *and* passes verification, it is named and thereafter emitted by the planner as a single atomic generator of $\mathfrak{D}$ —this is $\delta \mathfrak{D}$ made concrete. The promotion gate is VERIFYOP (Section 12): a composite becomes a generator only if it survives the external oracle.

10 The operad and parallel scheduling

Intuition. A reasoning act is a small program: a directed acyclic graph (DAG) of operators, some depending on others. Independent operators can run at once; the scheduler slices the DAG into layers of mutually independent work sized to the real machine.

Definition 10.1 (Operadic foliation). A reasoning act is a DAG $T = (N, \prec)$ of operator nodes. The scheduler foliates T into the antichain layers of the partial order $\prec$: layer 0 is the set of minimal nodes (in-degree 0), and layer $\ell + 1$ is the set of nodes all of whose predecessors lie in layers $\leq \ell$. Each layer is a *maximal commuting slice* executed in parallel; correctness follows because nodes in one antichain share no $\prec$ -dependency. Node “support” sets index which fine-complex vertices an operator reads/writes, so slices with disjoint support are safe to co-execute.

11 The lift–project adjunction $\mathcal{L} \dashv \mathcal{M}$

Intuition. Two worlds: free-form natural language, and structured reasoning DAGs. One map *lifts* language into a validated DAG; a partner map *projects* a DAG back into language. Meaning that carries no reasoning content collapses under the lift—it lands in the kernel.

Construction 11.1 (The boundary morphisms). Let π (“lift”, $\mathcal{L}$) send a natural-language request to a validated reasoning DAG, and G (“project”, $\mathcal{M}$) send a DAG to a natural-language rendering. Empirically, semantically void input maps into $\ker \pi$: a joke embedded in a mathematical prompt produced *zero* DAG nodes, exactly as a kernel element should. The lift is guarded by a validator that rejects malformed DAGs at the boundary (duplicate ids, dangling children, cycles by DFS colouring, absence of a terminal output node, disconnection), so the downstream engine only ever receives well-formed programs.

Claims and non-claims. $\mathcal{L} \dashv \mathcal{M}$ is the intended categorical *framing* (lift left-adjoint to project) and it organises the design usefully; the unit/counit triangle identities are *not* formally verified here, so “adjunction” should be read as a structuring principle plus concrete morphisms (π, G) with the observed $\ker \pi$ behaviour—not as a proved theorem. Stating this is part of the honesty invariant.

12 Verification: the external oracle

Intuition. Coherence governs whether to keep exploring or stop and reconcile; only an external checker touches *truth*. And “I could not check” must never be mistaken for “it is false.”

Definition 12.1 (Three-valued verification). VERIFYOP calls a sandboxed symbolic oracle returning one of three verdicts, which act on an obstruction counter Ω :

exit	verdict	effect on state
0	VERIFIED	obstruction cleared; state may crystallise
1	REFUTED	$\Omega += 1000$; forced into RESOLVE
2	UNVERIFIABLE	Ω <i>untouched</i>

Remark 12.2 (Absence of proof is not proof of absence). An earlier version treated every nonzero exit as failure, conflating REFUTED and UNVERIFIABLE; that would spike conflict over the oracle’s own blind spots and lock the system into RESOLVE over perfectly good mathematics. A checker crash also returns UNVERIFIABLE. This three-valued logic is the promotion gate of Construction 9.4.

13 The controller

Definition 13.1 (Coherence-gated mode). With threshold ε , the controller selects

$$\text{mode}(x) = \begin{cases} \text{RESOLVE}, & \rho > \varepsilon \text{ (reconcile the guilty edge before expanding),} \\ \text{EXPLORE}, & \rho \leq \varepsilon, \\ \text{UNKNOWN}, & \rho = \perp \text{ (no measurement).} \end{cases} \quad (16)$$

Remark 13.2 (Three modes, and empirical ε). The UNKNOWN branch is essential: when ρ is undefined the controller *refuses* to default to EXPLORE, because “no measurement” is not “all is well.” The threshold is calibrated, not guessed: on real 384-dimensional embeddings, agreement gave $\rho \approx 0.00$, related-but-distinct ≈ 0.06 , and off-topic ≈ 0.14 , so $\varepsilon \approx 0.10$ is a principled starting gate. Two cautions from live runs: ρ is *not* monotonic in the number of active organs (adding a well-aligned organ raises the normaliser faster than the discord, so ρ dilutes), and comparisons across differently-sized complexes must be made with care.

14 What is and is not claimed

Intuition. The topology’s deepest gift is not that it makes the system smart, but that it makes the claim of augmentation *falsifiable*: memory, consistency, and growth become operations on *one* object, with cheap provable guarantees and a live measure of whether augmentation is actually happening.

Diagnostic vs. constitutive use. A structure can *measure* a process happening elsewhere (diagnostic) or *be part of* the computation (constitutive). Today the sheaf is used diagnostically: the LLM reasons, ρ grades coherence. Four identified moves promote it toward constitutive use, none requiring a smarter LLM:

- (1) *sheaf diffusion as inference*: running $\dot{x} = -Lx$ to equilibrium converges to the orthogonal projection onto $\ker L = H^0$ —it *constructs* the least-discord global section, a reconciled state no single organ held (with identity restrictions this is consensus/averaging; its power grows with non-trivial maps);
- (2) *non-trivial (learned) restriction maps*, the formal home of “quality of a connection” (cf. Prop. 4.4 box);
- (3) *higher-cell Laplacians*, propagating joint constraints pairwise message-passing cannot represent;
- (4) *persistence as structural learning* (Construction 9.4).

The creativity boundary (Boden). With frozen LLM weights the system realises *combinatorial* and *exploratory* creativity—novel recombination, and search within a fixed representational space—at full strength. *Transformational* creativity (enlarging the space itself) is largely out of reach, with one genuine sliver: each promoted operator ($\delta\mathfrak{D}$) enlarges the system’s *effective* action space. So universal adaptivity is a claim about *organisation*, bounded by the model’s expressible *content*.

Claims and non-claims. Bottom line. The simplices do not hand you augmented intelligence. They hand you a rigorous, measurable, self-checking substrate on which augmentation-via-structure *might* emerge—through measured cross-organ error-correction and accumulation—and on which you can *tell whether it is emerging rather than deceiving yourself*. Whether the

bet wins (whole > sum) is unknown, and no elegance of topology guarantees it. The instrumentation is what makes it a real research bet rather than a beautiful story.

15 Outlook: predictive-coding sheaf diffusion (proposed, unimplemented)

Claims and non-claims. Everything in this section is a proposal, not a result. Nothing here has been implemented or numerically verified; it is recorded now, in the same implementation-free register as the rest of the paper, because the identities below were *derived* (not guessed) and are the concrete answer to “how does this architecture move from measuring cognition to constituting it.” Status: endorsed as a plan, work not yet started.

Intuition. Section 14 named the move: stop only *scoring* agreement between organs: let the structure *compute* with it. Four principles from computational neuroscience turn out to be the same four moves needed to do that, each attacking one of the architecture’s current hand-set magic numbers.

15.1 Predictive coding is precision-weighted sheaf diffusion

Proposition 15.1 (The identity). *Let each edge carry a scalar precision $\pi_e > 0$ (the confidence attached to that organ pair’s agreement). Define the predictive-coding free energy*

$$\mathcal{F}(x) = \sum_{e=\{u,v\}} \pi_e \|\varepsilon_e\|^2, \quad \varepsilon_e = (\delta x)_e = F_{u \leq e}(x_u) - F_{v \leq e}(x_v). \quad (17)$$

Then $\mathcal{F}$ is exactly the discord ω of Definition 4.2 with edge weights π_e : $\mathcal{F}(x) = x^\top L_\pi x$ for the precision-weighted Laplacian $L_\pi = \delta^\top \Pi \delta$, $\Pi = \text{diag}(\pi_e)$. Inference (settling on a reconciled belief) is the gradient flow $\dot{x} = -\nabla \mathcal{F}(x) = -L_\pi x$, i.e. sheaf diffusion (item (1) of Section 14) with a precision-weighted Laplacian in place of the unweighted one.

Remark 15.2 (This is the fine-level flow, one level up). Construction 7.1’s RK4 descent $\dot{x} = -\nabla \Phi(x)$ already realises predictive-coding inference *inside* an organ’s fine complex. Proposition 15.1 proposes the same dynamics at the *coarse* level, on L_π , closing the diagnostic/constitutive gap identified in Section 14.

15.2 Learning the restriction maps at zero training cost

Claims and non-claims. Section 14 flagged non-trivial restriction maps as the formal home of “connection quality,” with the open question: learned from what, at zero budget, with no backpropagation infrastructure? Predictive coding answers this with a purely local rule.

Construction 15.3 (Oja-stabilised Hebbian update). Treat each restriction map $R_{v,e} := F_{v \leq e}$ as a learnable matrix. The predictive-coding gradient is local to the edge:

$$\frac{\partial \mathcal{F}}{\partial R_{u,e}} = 2\pi_e \varepsilon_e x_u^\top, \quad \Delta R_{u,e} = -\eta \pi_e \varepsilon_e x_u^\top \quad (18)$$

for a step size $\eta > 0$ — a rank-one Hebbian update using only quantities already local to organ u and edge e . Raw Hebbian growth is unbounded, so an Oja normalisation term $-\eta \|\varepsilon_e\|^2 R_{u,e}$ is added to keep $R_{u,e}$ bounded, exactly as in Oja’s rule for Hebbian PCA. No backpropagation graph, no global loss: each edge updates from its own residual.

15.3 Criticality: a self-tuned threshold instead of a guessed one

Intuition. The coherence threshold $\varepsilon \approx 0.10$ (Section 13) was fit empirically to a handful of observations. Criticality theory offers a principled replacement: tune the system to sit where a single perturbation triggers, on average, exactly one further event.

Construction 15.4 (Branching-ratio setpoint). Define the branching ratio σ as the mean number of new binds triggered per co-activation event. Following Beggs–Plenz neuronal-avalanche theory and the Kinouchi–Copelli dynamic-range result, $\sigma \rightarrow 1$ (criticality) maximises the system’s dynamic range and marks the transition between over-damped (nothing restructures) and runaway (everything restructures) regimes. **Proposed:** replace the fixed ε with a controller that adjusts it to hold $\sigma \approx 1$, measured where sample size is large enough to mean anything (the fine level; the six-organ coarse level uses an empirical-quantile ε meanwhile).

15.4 Homeostatic growth trigger

Construction 15.5 (Deviation-from-setpoint trigger). Following Butz–van Ooyen structural plasticity, let $a(t)$ be a leaky-integrated activity/coherence signal with target a^* ; growth (attachment, Section 9.4) is triggered by *integrated* deviation,

$$\dot{z} = \nu(a^* - a(t)), \quad (19)$$

gating attachment only once z crosses a bound. This is thrash-resistant by construction: a single transient coherence spike does not trigger growth, only a sustained deviation does — replacing the fixed decay constant of Section 9 with an integrator.

15.5 Selectionism: variation, verification, promotion

Construction 15.6 (Attachment as selection). When diffusion (Proposition 15.1) fails to reduce $\mathcal{F}$ below tolerance, generate k candidate attachments at the located obstruction (the worst edge of Remark 4.7, or its cohomological generalisation, the harmonic representative of a persisting $[\zeta] \in H^1$); each candidate is an Edelman-style *variant*. VERIFYOP (Section 12) is the *sole* selection criterion: a variant is promoted to $\mathfrak{D}$ (Construction 9.4) only if it survives verification. Coherence improvement $\Delta\rho$ alone must never promote a variant — a fluent but false reconciliation is exactly the failure mode this excludes.

Claims and non-claims. What this layer does and does not buy, stated plainly.

- It does *not* raise the LLM’s per-call reasoning quality; the k attachment samples are *aimed and rationed* by the located obstruction, not a smarter model.
- Diffusion reconciles existing information; it never manufactures a true fact. VERIFYOP remains the sole arbiter of truth, unconditionally.
- All four mechanisms are *slope* mechanisms: on day one, with $\pi_e \equiv 1$, fixed ε , a reactive (non-integrated) trigger, and $k = 1$, the system degrades exactly to the architecture described in Sections 2–13. The bet, as always, is on accumulation, not on an improved first day.
- Criticality is used here as an *engineering* target (maximise dynamic range), not asserted as a governing law of cognition; it is meaningful only at sample sizes where a branching ratio is statistically defined.

The one-sentence summary: promote the coherence measure from a *scorer* to a *search controller* — detect struggle as a persisting obstruction, read the obstruction as the address of the needed

structure, spend the scarce LLM sample exactly there, verify, and promote survivors, with the same rule applying self-similarly at every level of the stratified complex.

A The driving application: a thermal Hodge spectrum on $\mathbb{CP}^3$

Intuition. The architecture is aimed at a concrete open problem, so the mathematics has a target, not just a form.

The target is the spectrum and time-evolution of a *thermally-weighted* Hodge–de Rham Laplacian Δ_β on 1-forms over complex projective space $\mathbb{CP}^3$ with the Fubini–Study metric, where the weight $e^{-\beta E}$ is supplied by a two-qubit Hamiltonian K . Two halves:

- *Retrievable core.* The *unweighted* Hodge spectrum on $\mathbb{CP}^n$ is classical (Ikeda–Taniguchi via $SU(n+1)$ representation theory [11]); the weighted operator Δ_β is a Witten / Bakry–Émery drift-Laplacian, a known deformation.
- *Novel frontier.* The thermally-weighted, symmetry-broken spectrum is a genuine new step and carries real confident-error risk.

The role of TATIANA on this problem is not to out-reason a frontier model in one shot but to *run the loop*: retrieve the classical core, grow concept-nodes, register a coherence spike at the frontier, verify what it can, and produce a grounded, honestly-hedged attempt—measuring its own augmentation as it goes. This is precisely the setting where the diagnostic hierarchy of Section 8.2 earns its keep: a lone LLM answer to the frontier half is confidently unfalsifiable; here the disagreement is measured and localised.

B Notation

symbol	meaning
$\mathcal{M} = (C, F, s)$	cognitive state space: complex, sheaf, current state
K, C_v	coarse (organ) complex; fine (concept) complex of organ v
$w(\sigma, t) \in [0, 1]$	coupling weight of coalition σ at time t
$F(\sigma)$	stalk on cell σ (here $\mathbb{R}^d$, $d = 384$)
$F_{v \triangleleft e}$	restriction map, vertex v into edge e
$\delta, L = \delta^\top \delta$	coboundary; sheaf Laplacian (symmetric PSD)
$\omega = x^\top Lx$	discord (Dirichlet energy)
$B, \rho = \omega / (B \ X\ _F^2)$	Anderson–Morley normaliser; coherence score $\in [0, 1]$
$H^0 = \ker L$	global sections (“coherent thoughts”)
H^1, b_1, b_0	gluing failure; structural hole; components
π_v	precision-weighted coarse-graining of organ v
$\Sigma_i = U_i U_i^\top + D_i I$	concept covariance (low-rank + isotropic)
W_2	Bures–Wasserstein distance between Gaussians
$\Phi, \dot{x} = -\nabla \Phi$	fine conflict functional; deductive gradient flow
$\mathfrak{D}, \delta \mathfrak{D}$	operator algebra; its growth (microneurons)
Ω	verification obstruction counter
ε	controller coherence threshold (≈ 0.10)
$\perp$	“unknown / undefined” (never silently 0)

C Cost and calibration summary

quantity	cost	note
ω, ρ	$O(E d)$	no eigensolver (Anderson–Morley)
worst edge, b_0	$O(E)$	per-edge sum; union-find
π_v isotropic	$O(nd)$	exact scalar precision average
π_v general	$O(ndk + d^3)$	Woodbury; verified to 2.2×10^{-16}
W_2 (isotropic Σ_1)	$O(dk^2 + k^3)$	$k \times k$ eigendecomposition only
paraphrase variance J	$k \times$ LLM calls	rationed to high-stakes
d	384	fixed embedding dimension
ε	≈ 0.10	calibrated on real embeddings

References

- [1] J. Curry, *Sheaves, Cosheaves and Applications*, Ph.D. thesis, University of Pennsylvania, 2014.
- [2] J. Hansen and R. Ghrist, “Toward a spectral theory of cellular sheaves,” *Journal of Applied and Computational Topology* **3** (2019), 315–358.
- [3] J. Hansen and R. Ghrist, “Opinion dynamics on discourse sheaves,” *SIAM J. Applied Mathematics* **81** (2021), 2033–2060.
- [4] W. N. Anderson and T. D. Morley, “Eigenvalues of the Laplacian of a graph,” *Linear and Multilinear Algebra* **18** (1985), 141–145.
- [5] C. Bodnar, F. Di Giovanni, B. Chamberlain, P. Liò, M. Bronstein, “Neural sheaf diffusion: a topological perspective on heterophily and oversmoothing in GNNs,” *NeurIPS*, 2022.
- [6] C. Bodnar et al., “Weisfeiler and Lehman go cellular: CW networks,” *NeurIPS*, 2021.
- [7] L.-H. Lim, “Hodge Laplacians on graphs,” *SIAM Review* **62** (2020), 685–715.
- [8] S. Abramsky and A. Brandenburger, “The sheaf-theoretic structure of non-locality and contextuality,” *New Journal of Physics* **13** (2011), 113036.
- [9] R. Bhatia, T. Jain, Y. Lim, “On the Bures–Wasserstein distance between positive definite matrices,” *Expositiones Mathematicae* **37** (2019), 165–191.
- [10] M. Boden, *The Creative Mind: Myths and Mechanisms*, 2nd ed., Routledge, 2004.
- [11] A. Ikeda and Y. Taniguchi, “Spectra and eigenforms of the Laplacian on S^n and $P^n(\mathbb{C})$,” *Osaka J. Mathematics* **15** (1978), 515–546.